

Question ID: 2e92cc21

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

The number a is 110% greater than the number b . The number b is 90% less than 47. What is the value of a ?

Correct Answer: 9.87, 987/100

Rationale

The correct answer is 9.87. It's given that the number a is 110% greater than the number b . It follows that $a = (1 + \frac{110}{100})b$, or $a = 2.1b$. It's also given that the number b is 90% less than 47. It follows that $b = (1 - \frac{90}{100})(47)$, or $b = 0.1(47)$, which yields $b = 4.7$. Substituting 4.7 for b in the equation $a = 2.1b$ yields $a = 2.1(4.7)$, which is equivalent to $a = 9.87$. Therefore, the value of a is 9.87.